

Package ‘statGraph’

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Imports igraph, MASS, rARPACK, cluster, foreach, parallel, doParallel, methods

Description Contains statistical methods to analyze graphs, such as graph parameter estimation, model selection based on the Graph Information Criterion, statistical tests to discriminate two or more populations of graphs, correlation between graphs, and clustering of graphs.
References: Takahashi et al. (2012) <[doi:10.1371/journal.pone.0049949](https://doi.org/10.1371/journal.pone.0049949)>, Fujita et al. (2017) <[doi:10.3389/fnins.2017.00066](https://doi.org/10.3389/fnins.2017.00066)>, Fujita et al. (2017) <[doi:10.1016/j.csda.2016.11.016](https://doi.org/10.1016/j.csda.2016.11.016)>, Tang et al. (2017) <[doi:10.3150/15-BEJ789](https://doi.org/10.3150/15-BEJ789)>, Tang et al. (2017) <[doi:10.1080/10618600.2016.1193505](https://doi.org/10.1080/10618600.2016.1193505)>, Ghoshdastidar et al. (2017) <[arXiv:1705.06168](https://arxiv.org/abs/1705.06168)>, Ghoshdastidar et al. (2017) <[arXiv:1707.00833](https://arxiv.org/abs/1707.00833)>, Cerqueira et al. (2017) <[doi:10.1109/TNSE.2017.2674026](https://doi.org/10.1109/TNSE.2017.2674026)>, Fraiman and Fraiman (2018) <[doi:10.1038/s41598-018-23152-5](https://doi.org/10.1038/s41598-018-23152-5)>, Fujita et al. (2019) <[doi:10.1093/comnet/cnz028](https://doi.org/10.1093/comnet/cnz028)>.

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anogva	<i>ANOGVA Analysis Of Graph Variability</i>
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Description

anogva statistically tests whether two or more sets of graphs are generated by the same random graph model. It is a generalization of the takahashi.test function.

Usage

anogva(Graphs, labels, maxBoot = 1000, dist = "KL", ...)

Arguments

- | | |
|---------|---|
| Graphs | a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues , such values will be used to compute their spectral density. |
| labels | an array of integers indicating the labels of each graph. |
| maxBoot | integer indicating the number of bootstrap resamplings (default 1000). |

`dist` string indicating if you want to use the "KL" (default), "JS" , "L1" or "L2" distances. "KL" means Kullback-Leibler divergence. "JS" means Jensen-Shannon divergence.

`...` Other relevant parameters for [graph.spectral.density](#).

Value

A list containing:

- `statistic`: the statistic of the test.
- `p.value`: the p-value of the test.

References

- Fujita, A., Vidal, M. C. and Takahashi, D. Y. (2017) A Statistical Method to Distinguish Functional Brain Networks. *_Front. Neurosci._*, *11*, 66. doi:10.3389/fnins.2017.00066.
- Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *_PLoS ONE_*, *7*, e49949. doi:10.1371/journal.pone.0049949.
- Silverman, B. W. (1986) *_Density Estimation_*. London: Chapman and Hall.
- Sturges, H. A. The Choice of a Class Interval. *_J. Am. Statist. Assoc._*, *21*, 65-66.
- Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *_Journal of the Royal Statistical Society series B_*, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
set.seed(1)
g1 <- g2 <- g3 <- list()
for (i in 1:20) {
  g1[[i]] <- igraph::sample_gnp(50, 0.50)
  g2[[i]] <- igraph::sample_gnp(50, 0.50)
  g3[[i]] <- igraph::sample_gnp(50, 0.52)
}
G <- c(g1, g2, g3)
label <- c(rep(1,20),rep(2,20),rep(3,20))
result <- anogva(G, label, maxBoot=50)
result
```

GIC

Graph Information Criterion (GIC)

Description

GIC returns the Kullback-Leibler divergence, L1 or L2 distance between an undirected graph and a given graph model using the exact or degree-based spectral densities.

Usage

```
GIC(Graph, model, p = NULL, dist = "KL", ...)
```

Arguments

Graph	the undirected graph (igraph object). If Graph has the attribute eigenvalues containing the eigenvalues of Graph, such values will be used to compute its spectral density.
model	either a list, a string, or a function describing a graph model: A list that represents the spectral density of a model. It contains the components "x" and "y", which are numeric vectors of the same size. The "x" component contains the points at which the density was computed and the "y" component contains the observed density. A string that indicates one of the following models: "ER" (Erdos-Renyi random graph), "GRG" (geometric random graph), "KR" (k regular random graph), "WS" (Watts-Strogatz model), and "BA" (Barabási-Albert model). When the argument model is a string, the user must also provide the model parameter by using the argument p. A function that returns a graph (igraph object) generated by a graph model. It must contain two arguments: the first one corresponds to the graph size and the second to the parameter of the model. The model parameter will be provided by the argument p of the GIC function.
p	the model parameter. The user must provide a valid parameter if the argument model is a string or a function. For the predefined models ("ER", "GRG", "KR", "WS", and "BA"), the parameter the probability to connect a pair of vertices, for the "ER" model (Erdos-Renyi random graph); the radius used to construct the geometric graph in a unit square, for the "GRG" model (geometric random graph); the degree k of a regular graph, for the "KR" model (k regular random graph); the probability to reconnect a vertex, for the "WS" model (Watts-Strogatz model); and the scaling exponent, for the "BA" model (Barabási-Albert model).
dist	string indicating if you want to use the "KL" (default), "L1" or "L2" distances. "KL" means Kullback-Leibler divergence.
...	Other relevant parameters for graph.spectral.density .

Value

A real number corresponding to the Kullback-Leibler divergence, L1 or L2 distance between the observed graph and the graph model.

References

Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *_PLoS ONE_*, *7*, e49949. doi:10.1371/journal.pone.0049949.

Silverman, B. W. (1986) *_Density Estimation_*. London: Chapman and Hall.

Sturges, H. A. The Choice of a Class Interval. _J. Am. Statist. Assoc._, *21*, 65-66.

Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. _Journal of the Royal Statistical Society series B_, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
set.seed(1)
G <- igraph::sample_gnp(n=50, p=0.5)

# Using a string to indicate the graph model
result1 <- GIC(G, "ER", 0.5)
result1

# Using a function to describe the graph model
# Erdos-Renyi graph
model <- function(n, p) {
  return (igraph::sample_gnp(n, p))
}
result2 <- GIC(G, model, 0.5)
result2
```

graph.acf

Auto Correlation Function Estimation for Graphs

Description

The function `graph.acf` computes estimates of the autocorrelation function for graphs.

Usage

```
graph.acf(Graphs, plot = TRUE)
```

Arguments

<code>Graphs</code>	a list of undirected graphs. If each graph has the attribute <code>eigenvalues</code> containing its eigenvalues, such values will be used to compute their spectral density.
<code>plot</code>	logical. If TRUE (default) the <code>graph.acf</code> is plotted.

Value

An object of class `acf`.

References

Fujita, A., Takahashi, D. Y., Balardin, J. B., Vidal, M. C. and Sato, J. R. (2017) Correlation between graphs with an application to brain network analysis. _Computational Statistics & Data Analysis_ *109*, 76-92.

Examples

```
set.seed(1)
G <- list()
p <- array(0, 100)
p[1:3] <- rnorm(3)
for (t in 4:100) {
  p[t] <- 0.5*p[t-3] + rnorm(1)
}
ma <- max(p)
mi <- min(p)
p <- (p - mi)/(ma-mi)
for (t in 1:100) {
  G[[t]] <- igraph::sample_gnp(100, p[t])
}
graph.acf(G, plot=TRUE)
```

graph.cem

Clustering Expectation-Maximization for Graphs (graph.cem)

Description

graph.cem clusters graphs following an expectation-maximization algorithm based on the Kullback-Leibler divergence between the spectral densities of the graph and of the random graph model.

Usage

```
graph.cem(Graphs, model, k, max_iter = 10, ...)
```

Arguments

Graphs	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
model	a string that indicates one of the following random graph models: "ER" (Erdos-Renyi random graph), "GRG" (geometric random graph), "KR" (k regular graph), "WS" (Watts-Strogatz model), and "BA" (Barabási-Albert model).
k	an integer specifying the number of clusters.
max_iter	the maximum number of expectation-maximization steps to execute.
...	Other relevant parameters for graph.param.estimator .

Value

a list containing three fields:

- labels: a vector of the same length of Graphs containing the clusterization labels.
- parameters: a vector containing the estimated parameters for the groups. It has the length equals to k.

References

Celeux, Gilles, and Gerard Govaert. "Gaussian parsimonious clustering models." *Pattern recognition* 28.5 (1995): 781-793.

Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *Journal of the Royal Statistical Society series B*, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
set.seed(42)
g <- list()
for(i in 1:2){
  g[[i]] <- igraph::sample_gnp(n=10, p=0.5)
}
for(i in 3:4){
  g[[i]] <- igraph::sample_gnp(n=10, p=1)
}
res <- graph.cem(g, model="ER", k=2, max_iter=1)
res
```

graph.cerqueira.test *Cerqueira's hypothesis testing for random graphs.*

Description

Andressa Cerqueira, Daniel Fraiman, Claudia D. Vargas and Florencia Leonardi non-parametric test of hypotheses to verify if two samples of random graphs were originated from the same probability distribution.

Usage

```
graph.cerqueira.test(
  Graphs1,
  Graphs2,
  maxPer = 300,
  alpha = 0.05,
  printResult = FALSE
)
```

Arguments

Graphs1	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
Graphs2	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
maxPer	integer indicating the number of bootstrap resamples (default is 300).
alpha	the significance level for the test (default is 0.05).
printResult	logical indicating if the test must print the result (default is FALSE).

Details

Given two identically independently distributed (idd) samples of graphs Graphs1 and Graphs2, the test verifies if they have the same distribution by calculating the mean distance D from Graphs1 to Graphs2. The test rejects the null hypothesis if D is greater than the $(1-\alpha)$ -quantile of the distribution of the test under the null hypothesis.

Value

A list containing:

- `test_stats`: the value of the test.
- `p_value`: the p-value of the test.
- `reject_threshold`: The $1-\alpha$ -quantile of the test distribution under the null hypothesis.
- `bootstrap_samples`: The test distribution on the bootstrap resamples.

References

Andressa Cerqueira, Daniel Fraiman, Claudia D. Vargas and Florencia Leonardi. "A test of hypotheses for random graph distributions built from EEG data", <https://ieeexplore.ieee.org/document/7862892>

Examples

```
require(igraph)
set.seed(42)

## test under H0
Graphs1 <- Graphs2 <- list()
for(i in 1:10){
  Graphs1[[i]] <- igraph::erdos.renyi.game(50,0.5)
  Graphs2[[i]] <- igraph::erdos.renyi.game(50,0.5)
}
k <- graph.cerqueira.test(Graphs1, Graphs2, printResult = TRUE)

## test under H1
Graphs1 <- Graphs2 <- list()
for(i in 1:10){
  Graphs1[[i]] <- igraph::erdos.renyi.game(50,0.5)
  Graphs2[[i]] <- igraph::erdos.renyi.game(50,0.6)
}
k <- graph.cerqueira.test(Graphs1, Graphs2, printResult = TRUE)
```

graph.cor.test	<i>Test for Association / Correlation Between Paired Samples of Graphs</i>
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Description

graph.cor.test tests for association between paired samples of graphs, using Spearman's rho correlation coefficient.

Usage

```
graph.cor.test(Graphs1, Graphs2)
```

Arguments

Graphs1	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
Graphs2	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.

Value

- statistic: the value of the test statistic.
- p.value: the p-value of the test.
- estimate: the estimated measure of association rho.

References

Fujita, A., Takahashi, D. Y., Balardin, J. B., Vidal, M. C. and Sato, J. R. (2017) Correlation between graphs with an application to brain network analysis. *_Computational Statistics & Data Analysis_* *109*, 76-92.

Examples

```
set.seed(1)
G1 <- G2 <- list()

p <- MASS::mvrnorm(50, mu=c(0,0), Sigma=matrix(c(1, 0.5, 0.5, 1), 2, 2))

ma <- max(p)
mi <- min(p)
p[,1] <- (p[,1] - mi)/(ma - mi)
p[,2] <- (p[,2] - mi)/(ma - mi)

for (i in 1:50) {
  G1[[i]] <- igraph::sample_gnp(50, p[i,1])
  G2[[i]] <- igraph::sample_gnp(50, p[i,2])
}
graph.cor.test(G1, G2)
```

graph.entropy	<i>Graph spectral entropy</i>
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Description

graph.entropy returns the spectral entropy of an undirected graph.

Usage

```
graph.entropy(Graph, ...)
```

Arguments

Graph	the undirected graph (igraph object). If Graph has the attribute eigenvalues containing the eigenvalues of Graph, such values will be used to compute its spectral density.
...	Other relevant parameters for graph.spectral.density .

Value

a real number corresponding to the graph spectral entropy.

References

Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *_PLoS ONE_, *7*, e49949*. doi:10.1371/journal.pone.0049949.

Examples

```
G <- igraph::sample_gnp(n=100, p=0.5)
entropy <- graph.entropy(Graph = G)
entropy
```

graph.fraiman.test	<i>Fraiman's hypothesis testing for random graphs.</i>
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Description

Daniel Fraiman and Ricardo Fraiman test for network differences between groups with an analysis of variance test (ANOVA).

Usage

```
graph.fraiman.test(Graphs, maxPer = 300, alpha = 0.05, printResult = FALSE)
```

Arguments

Graphs	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
maxPer	integer indicating the number of bootstrap resamples (default is 300).
alpha	the significance level for the test (default is 0.05).
printResult	logical indicating if the test must print the result (default is FALSE).

Details

Given a list of graphs, the test verifies if all the subpopulations have the same mean network, under the alternative that at least one subpopulation has a different mean network.

Value

A list containing:

- test_stats: the value of the test.
- p_value: the p-value of the test.
- bootstrap_samples: the test distribution on the bootstrap resamples.

References

Fraiman, Daniel, and Ricardo Fraiman. "An ANOVA approach for statistical comparisons of brain networks", <https://www.nature.com/articles/s41598-018-23152-5>

Examples

```
set.seed(42)

## test under H0
a <- b <- list()
for(i in 1:10){
  a[[i]] <- igraph::erdos.renyi.game(50,0.5)
  b[[i]] <- igraph::erdos.renyi.game(50,0.5)
}
Graphs <- list(a,b)
k <- graph.fraiman.test(Graphs, printResult = TRUE)

## test under H1
a <- b <- list()
for(i in 1:10){
  a[[i]] <- igraph::erdos.renyi.game(50,0.5)
  b[[i]] <- igraph::erdos.renyi.game(50,0.6)
}
Graphs <- list(a,b)
k <- graph.fraiman.test(Graphs, printResult = TRUE)
```

```
graph.ghoshdastidar.test
```

Ghoshdastidar hypothesis testing for large random graphs.

Description

Given two lists of graphs generated by the inhomogeneous random graph model, ghoshdastidar tests if they were generated by the same parameters.

Usage

```
graph.ghoshdastidar.test(
  Graphs1,
  Graphs2,
  maxPer = 300,
  alpha = 0.05,
  two.sample = FALSE,
  printResult = FALSE
)
```

Arguments

Graphs1	a list of undirected Graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
Graphs2	a list of undirected Graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
maxPer	integer indicating the number of bootstrap resamples (default is 300).
alpha	the significance level for the test (default is 0.05).
two.sample	logical. If TRUE the sets contain only one graph each. If FALSE the sets contain more than one graph each (default is FALSE).
printResult	logical indicating if the test must print the result (default is FALSE).

Value

A list containing:

- `test_stats`: the value of the test.
- `p_value`: the p-value of the test (only returned when the parameter `two.sample` is FALSE).
- `bootstrap_samples`: the test distribution on the bootstrap resamples (only returned when the parameter `two.sample` is FALSE).

References

Ghoshdastidar, Debarghya, et al. "Two-sample tests for large random graphs using network statistics". arXiv preprint arXiv:1705.06168 (2017).

Ghoshdastidar, Debarghya, et al. "Two-sample hypothesis testing for inhomogeneous random graphs". arXiv preprint, arXiv:1707.00833 (2017).

Examples

```
set.seed(42)

## test for sets with only one graph each under H0
Graphs1 <- Graphs2 <- list()
Graphs1[[1]] <- igraph::erdos.renyi.game(300, 0.6)
Graphs2[[1]] <- igraph::erdos.renyi.game(300, 0.6)
D <- graph.ghoshdastidar.test(Graphs1, Graphs2, two.sample= TRUE, printResult = TRUE)

## test for sets with only one graph each under H1
Graphs1 <- Graphs2 <- list()
Graphs1[[1]] <- igraph::erdos.renyi.game(300, 0.6)
Graphs2[[1]] <- igraph::erdos.renyi.game(300, 0.7)
D <- graph.ghoshdastidar.test(Graphs1, Graphs2, two.sample= TRUE, printResult = TRUE)
```

graph.hclust

Hierarchical cluster analysis on a list of graphs.

Description

Given a list of graphs, `graph.hclust` builds a hierarchy of clusters according to the Jensen-Shannon divergence between graphs.

Usage

```
graph.hclust(Graphs, k, clus_method = "complete", dist = "JS", ...)
```

Arguments

<code>Graphs</code>	a list of undirected graphs. If each graph has the attribute <code>eigenvalues</code> containing its eigenvalues, such values will be used to compute their spectral density.
<code>k</code>	the number of clusters.
<code>clus_method</code>	the agglomeration method to be used. This should be (an unambiguous abbreviation of) one of <code>"ward.D"</code> , <code>"ward.D2"</code> , <code>"single"</code> , <code>"complete"</code> , <code>"average"</code> (= UPGMA), <code>"mcquitty"</code> (= WPGMA), <code>"median"</code> (= WPGMC) or <code>"centroid"</code> (= UPGMC).
<code>dist</code>	string indicating if you want to use the <code>"JS"</code> (default), <code>"L1"</code> or <code>"L2"</code> distances. <code>"JS"</code> means Jensen-Shannon divergence.
<code>...</code>	Other relevant parameters for graph.spectral.density .

Value

A list containing:

- `hclust`: an object of class `hclust` which describes the tree produced by the clustering process.
- `cluster`: the clustering labels for each graph.

References

- Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *_PLOS ONE_*, *7*, e49949. doi:10.1371/journal.pone.0049949.
- Silverman, B. W. (1986) *_Density Estimation_*. London: Chapman and Hall.
- Sturges, H. A. The Choice of a Class Interval. *_J. Am. Statist. Assoc._*, *21*, 65-66.
- Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *_Journal of the Royal Statistical Society series B_*, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
set.seed(1)
G <- list()
for (i in 1:5) {
  G[[i]] <- igraph::sample_gnp(50, 0.5)
}
for (i in 6:10) {
  G[[i]] <- igraph::sample_smallworld(1, 50, 8, 0.2)
}
for (i in 11:15) {
  G[[i]] <- igraph::sample_pa(50, power = 1, directed = FALSE)
}
graph.hclust(G, 3)
```

graph.kmeans

K-means for Graphs

Description

graph.kmeans clusters graphs following a k-means algorithm based on the Jensen-Shannon divergence between the spectral densities of the graphs.

Usage

```
graph.kmeans(Graphs, k, nstart = 2, dist = "JS", ...)
```

Arguments

Graphs	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
k	an integer specifying the number of clusters.
nstart	the number of trials of k-means clusterizations. The algorithm returns the clusterization with the best silhouette.
dist	string indicating if you want to use the "JS" (default), "L1" or "L2" distances. "JS" means Jensen-Shannon divergence.
...	Other relevant parameters for graph.spectral.density .

Value

a vector of the same length of Graphs containing the clusterization labels.

References

MacQueen, James. "Some methods for classification and analysis of multivariate observations." Proceedings of the fifth Berkeley symposium on mathematical statistics and probability. Vol. 1. No. 14. 1967.

Lloyd, Stuart. "Least squares quantization in PCM." IEEE transactions on information theory 28.2 (1982): 129-137.

Examples

```
set.seed(42)
g <- list()
for(i in 1:5){
  g[[i]] <- igraph::sample_gnp(30, p=0.2)
}
for(i in 6:10){
  g[[i]] <- igraph::sample_gnp(30, p=0.5)
}
res <- graph.kmeans(g, k=2, nstart=2)
res
```

graph.model.selection *Graph model selection*

Description

graph.model.selection selects the graph model that best approximates the observed graph according to the Graph Information Criterion (GIC).

Usage

```
graph.model.selection(Graph, models = NULL, parameters = NULL, ...)
```

Arguments

Graph	the undirected graph (igraph object). If Graph has the attribute eigenvalues containing the eigenvalues of Graph, such values will be used to compute its spectral density.
models	either a vector of strings, or a list of functions: A vector of strings containing some of the following models: "ER" (Erdos-Renyi random graph), "GRG" (geometric random graph), "KR" (k regular random graph), "WS" (Watts-Strogatz model), and "BA" (Barabási-Albert model).

	A list of functions. Each function returns a graph (igraph object) generated by a graph model and has two arguments: the graph size and the model parameter, in this order.
	If the argument <code>models</code> is NULL, then the "ER", "WS", and "BA" models will be considered for the model selection.
<code>parameters</code>	list of numeric vectors or list of lists. If a list of numeric vectors is given, then each vector contains the values that will be considered for the parameter estimation of the corresponding model. If a list of lists is given, then each list contains <code>lo</code> and <code>hi</code> elements that indicate the model's parameter search interval <code><lo,hi></code> . If the user does not provide the argument <code>parameters</code> , then default values are used for the predefined models ("ER", "GRG", "KR", "WS", and "BA") as done in <code>graph.param.estimator</code> .
<code>...</code>	Other relevant parameters for <code>graph.param.estimator</code> .

Value

A list containing:

- `model`: the indice(s) or name(s) of the selected model(s), i. e. the model(s) that minimize(s) the Graph Information Criterion (GIC).
- `estimates`: a matrix in which each row corresponds to a model, the column "param" corresponds to the parameter estimate, and the column "GIC" corresponds to the Graph Information Criterion (GIC), i. e. the distance measure (Kullback-Leibler divergence by default) between the observed graph and the model.

References

- Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *PLoS ONE*, *7*, e49949. doi:10.1371/journal.pone.0049949.
- Silverman, B. W. (1986) *Density Estimation*. London: Chapman and Hall.
- Sturges, H. A. The Choice of a Class Interval. *J. Am. Statist. Assoc.*, *21*, 65-66.
- Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *Journal of the Royal Statistical Society series B*, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
## Example using an igraph object as input data
set.seed(1)
G <- igraph::sample_gnp(n=30, p=0.5)

# Using strings to indicate the graph models
result1 <- graph.model.selection(G, models=c("ER", "WS"), eps = 0.5)
result1

## Using functions to describe the graph models
# Erdos-Renyi graph
```

```

model1 <- function(n, p) {
  return(igraph::sample_gnp(n, p))
}
# Watts-Strogatz small-world graph
model2 <- function(n, pr, K=8) {
  return(igraph::sample_smallworld(1, n, K, pr))
}
parameters <- list(seq(0.01, 0.99, 0.49), seq(0.01, 0.99, 0.49))
result2 <- graph.model.selection(G, list(model1, model2), parameters)
result2

```

graph.mult.scaling *Multidimensional scaling of graphs*

Description

graph.mult.scaling performs multidimensional scaling of graphs. It takes the Jensen-Shannon divergence between graphs (JS) and uses the cmdscale function from the stats package to obtain a set of points such that the distances between the points are similar to JS.

Usage

```

graph.mult.scaling(
  Graphs,
  plot = TRUE,
  type = "n",
  dist = "JS",
  main = "",
  ...
)

```

Arguments

Graphs	a list of undirected graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
plot	logical. If TRUE (default) the points chosen to represent the Jensen-Shannon divergence between graphs are plotted.
type	what type of plot should be drawn. The default value is "n", which indicates that the points will not be plotted (i. e. only the labels of the graphs will be plotted).
dist	string indicating if you want to use the "JS" (default), "L1" or "L2" distances. "JS" means Jensen-Shannon divergence.
main	title of the plot (default value is "").
...	additional parameters for graph.spectral.density .

Value

A matrix in which each column corresponds to a coordinate and each row corresponds to a graph (point). Then, each row gives the coordinates of the points chosen to represent the Jensen-Shannon divergence between graphs.

References

- Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *PLoS ONE*, *7*, e49949. doi:10.1371/journal.pone.0049949.
- Silverman, B. W. (1986) *Density Estimation*. London: Chapman and Hall.
- Sturges, H. A. The Choice of a Class Interval. *J. Am. Statist. Assoc.*, *21*, 65-66.
- Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *Journal of the Royal Statistical Society series B*, *53*, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
set.seed(1)
G <- list()
for (i in 1:5) {
  G[[i]] <- igraph::sample_gnp(50, 0.5)
}
for (i in 6:10) {
  G[[i]] <- igraph::sample_smallworld(1, 50, 8, 0.2)
}
for (i in 11:15) {
  G[[i]] <- igraph::sample_pa(50, power = 1, directed = FALSE)
}
graph.mult.scaling(G)
```

graph.param.estimator *Graph parameter estimator*

Description

graph.param.estimator estimates the parameter that best approximates the model to the observed graph according to the Graph Information Criterion (GIC).

Usage

```
graph.param.estimator(
  Graph,
  model,
  interval = NULL,
  eps = 0.01,
  search = "grid",
  ...
)
```

Arguments

Graph	the undirected graph (igraph object). If Graph has the attribute 'eigenvalues' containing the eigenvalues of Graph, such values will be used to compute spectral density of the graph.
model	either a string or a function: A string that indicates one of the following models: "ER" (Erdos-Renyi random graph), "GRG" (geometric random graph), "KR" (k regular random graph), "WS" (Watts-Strogatz model), and "BA" (Barabási-Albert model). A function that returns a graph (represented by its adjacency matrix) generated by a graph model. It must contain two arguments: the first one corresponds to the graph size and the second to the parameter of the model.
interval	numeric vector containing the values that will be considered for the parameter estimation, or a list containing 'lo' and 'hi' that indicates the model's parameter search interval <lo,hi>. The graph.param.estimator will return the element of 'parameter' that minimizes the GIC. If the user does not provide the argument parameters, and model is a string, then default values are used for the predefined models ("ER", "GRG", "KR", "WS", and "BA"). The default parameter argument corresponds to a sequence from 0 to 1 with step eps for the "ER" model (Erdos-Renyi random graph), in which the parameter corresponds to the probability to connect a pair of vertices; 0 to sqrt(2) with step eps for the "GRG" model (geometric random graph), in which the parameter corresponds to the radius used to construct the geometric graph in a unit square; 0 to 'n' with step n*eps for the "KR" model (k regular random graph), in which the parameter of the model corresponds to the degree k of a regular graph; 0 to 1 with step eps for the "WS" model (Watts-Strogatz model), in which the parameter corresponds to the probability to reconnect a vertex; and 0 to 3 with step eps for the "BA" model (Barabási-Albert model), in which the parameter corresponds to the scaling exponent.
eps	precision of the grid and ternary search (default is 0.01).
search	string that indicates the search algorithm to find the parameter with the smallest GIC. If "grid" (default) parameter is estimated using grid search, and only works when method is not "fast". If "ternary" parameter is estimated using ternary search.
...	Other relevant parameters for GIC .

Value

A list containing:

- param: the parameter estimate. For the "ER", "GRG", "KR", "WS", and "BA" models, the parameter corresponds to the probability to connect a pair of vertices, the radius used to construct the geometric graph in a unit square, the degree k of a regular graph, the probability to reconnect a vertex, and the scaling exponent, respectively.
- dist: the distance between the observed graph and the graph model with the estimated parameter.

References

- Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *_PLOS ONE_*, *7*, e49949. doi:10.1371/journal.pone.0049949.
- Silverman, B. W. (1986) *_Density Estimation_*. London: Chapman and Hall.
- Sturges, H. A. The Choice of a Class Interval. *_J. Am. Statist. Assoc._*, *21*, 65-66.
- Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *_Journal of the Royal Statistical Society series B_*, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
set.seed(1)
G <- igraph::sample_gnp(n=50, p=0.5)

# Using a string to indicate the graph model
result1 <- graph.param.estimator(G, "ER", eps=0.25)
result1

# Using a function to describe the graph model
# Erdos-Renyi graph
set.seed(1)
model <- function(n, p) {
  return(igraph::sample_gnp(n, p))
}
result2 <- graph.param.estimator(G, model, seq(0.2, 0.8, 0.1))
result2
```

graph.spectral.density

Graph spectral density

Description

graph.spectral.density returns the exact or degree-based spectral density in the interval <from,to> by using npoints discretization points.

Usage

```
graph.spectral.density(Graph, method = "diag", ...)
```

Arguments

- | | |
|--------|--|
| Graph | the undirected graph (igraph object). If Graph has the attribute eigenvalues containing the eigenvalues of Graph, such values will be used to compute its spectral density. |
| method | String that specifies the method to obtain the spectral density. It can take two possible values 'diag' (Default) and 'fast'. If 'diag' is used then the exact spectral density is obtained, otherwise the degree-based spectral density is obtained. |
| ... | Other relevant parameters to obtain the spectral density such as from, to, an npoints. from, to, specify the lower and upper bound of the eigenvalues' support (automatically computed if not given); and npoints is the number of discretization points (default 1024) of the interval <from,to>. There are other parameters that depend on the value of the parameter method: <ul style="list-style-type: none"> • method= 'diag': then the parameter bandwidth can be used. This parameter is a string that specifies the criterion to choose the bandwidth during the spectral density estimation. Choose between the following criteria: "Silverman" (default), "Sturges", "bcv", "ucv" and "SJ". "bcv" is an abbreviation of biased cross-validation, while "ucv" means unbiased cross-validation. "SJ" implements the methods of Sheather & Jones (1991) to select the bandwidth using pilot estimation of derivatives. • method= 'fast': then the parameter numCores can be used. This parameter specifies the number of cores (default 1) to use for parallelization. |

Value

Returns the spectral density of the graph.

References

- #' Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *_PLoS ONE_*, *7*, e49949. doi:10.1371/journal.pone.0049949.
- Silverman, B. W. (1986) *_Density Estimation_*. London: Chapman and Hall.
- Sturges, H. A. The Choice of a Class Interval. *_J. Am. Statist. Assoc._*, *21*, 65-66.
- Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *_Journal of the Royal Statistical Society series B_*, 53, 683-690. <http://www.jstor.org/stable/2345597>.
- Newman, M. E. J., Zhang, X., & Nadakuditi, R. R. (2019). Spectra of random networks with arbitrary degrees. *Physical Review E*, 99(4), 042309.

Examples

```
set.seed(42)
G <- igraph::sample_smallworld(dim = 1, size = 50, nei = 2, p = 0.2)

# Obtain the spectral density
density <- graph.spectral.density(Graph = G)
density
```

graph.takahashi.test *Test for the Jensen-Shannon divergence between graphs*

Description

graph.takahashi.test tests whether two sets of graphs were generated by the same random graph model. This bootstrap test is based on the Jensen-Shannon (JS) divergence between graphs.

Usage

```
graph.takahashi.test(Graphs1, Graphs2, maxBoot = 1000, dist = "JS", ...)
```

Arguments

Graphs1	a list of undirected Graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
Graphs2	a list of undirected Graphs. If each graph has the attribute eigenvalues containing its eigenvalues, such values will be used to compute their spectral density.
maxBoot	integer indicating the number of bootstrap resamplings (default 1000).
dist	string indicating if you want to use the "JS" (default), "L1" or "L2" distances. "JS" means Jensen-Shannon divergence.
...	Other relevant parameters for graph.spectral.density .

Details

Given two lists of graphs, Graphs1 and Graphs2, graph.takahashi.test tests H0: "JS divergence between Graphs1 and Graphs2 is 0" against H1: "JS divergence between Graphs1 and Graphs2 is larger than 0".

Value

A list containing:

- dist: the distance between Graphs1 and Graphs2.
- p.value: the p-value of the test.

References

- Takahashi, D. Y., Sato, J. R., Ferreira, C. E. and Fujita A. (2012) Discriminating Different Classes of Biological Networks by Analyzing the Graph Spectra Distribution. *_PLOS ONE_*, *7*, e49949. doi:10.1371/journal.pone.0049949.
- Silverman, B. W. (1986) *_Density Estimation_*. London: Chapman and Hall.
- Sturges, H. A. The Choice of a Class Interval. *_J. Am. Statist. Assoc._*, *21*, 65-66.
- Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *_Journal of the Royal Statistical Society series B_*, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```

set.seed(1)
G1 <- G2 <- list()
for (i in 1:20) {
  G1[[i]] <- igraph::sample_gnp(n=50, p=0.5)
}
for (i in 1:20) {
  G2[[i]] <- igraph::sample_gnp(n=50, p=0.51)
}
result <- graph.takahashi.test(G1, G2, maxBoot=100)
result

```

graph.tang.test	<i>Tang hypothesis testing for random graphs.</i>
-----------------	---

Description

Given two independent finite-dimensional random dot product graphs, `graph.tang.test` tests if they have generating latent positions that are drawn from the same distribution.

Usage

```

graph.tang.test(
  Graph1,
  Graph2,
  dim,
  sigma = NULL,
  alpha = 0.05,
  bootstrap_sample = 200,
  printResult = FALSE
)

```

Arguments

Graph1	the first undirected graph (igraph object). If Graph has the attribute <code>eigenvalues</code> containing the eigenvalues of Graph, such values will be used to compute its spectral density.
Graph2	the second undirected graph (igraph object). If Graph has the attribute <code>eigenvalues</code> containing the eigenvalues of Graph, such values will be used to compute its spectral density.
dim	dimension of the adjacency spectral embedding.
sigma	a real value indicating the kernel bandwidth. If <code>NULL</code> (default) the bandwidth is calculated by the method.
alpha	the significance level for the test (default is <code>0.05</code>).
bootstrap_sample	integer indicating the number of bootstrap resamples (default is <code>200</code>).
printResult	logical indicating if the test must print the result (default is <code>FALSE</code>).

Value

A list containing:

- X1: the embedding of Graph1.
- X2: the embedding of Graph2.
- test_stats: the value of the test.
- p_value: the p-value of the test.
- bootstrap_samples: the test distribution on the bootstrap resamples.

References

Tang, Minh, et al. "A nonparametric two-sample hypothesis testing problem for random graphs." *Bernoulli* 23.3 (2017): 1599-1630.

Tang, Minh, et al. "A semiparametric two-sample hypothesis testing problem for random graphs." *Journal of Computational and Graphical Statistics* 26.2 (2017): 344-354.

Examples

```
set.seed(42)

## test under H0
lpvs <- matrix(rnorm(200), 20, 10)
lpvs <- apply(lpvs, 2, function(x) { return (abs(x)/sqrt(sum(x^2))) })
Graph1 <- igraph::sample_dot_product(lpvs)
Graph2 <- igraph::sample_dot_product(lpvs)
D <- graph.tang.test(Graph1, Graph2, 5, printResult = TRUE)

## test under H1
lpvs2 <- matrix(pnorm(200), 20, 10)
lpvs2 <- apply(lpvs2, 2, function(x) { return (abs(x)/sqrt(sum(x^2))) })
g2 <- suppressWarnings(igraph::sample_dot_product(lpvs2))
D <- graph.tang.test(Graph1, Graph2, 5, printResult = TRUE)
```

sp.anogva

Semi-Parametric Analysis Of Graph Variability (ANOGVA)

Description

sp.anogva statistically tests whether two or more graphs are generated by the same model and set of parameters.

Usage

```
sp.anogva(Graphs, model, maxBoot = 100, ...)
```

Arguments

Graphs	a list of undirected graphs. If each graph has the attribute <code>eigenvalues</code> containing its eigenvalues, such values will be used to compute their spectral density.
model	A string that indicates one of the following models: "ER" (Erdos-Renyi random graph model), "GRG" (geometric random graph model), "WS" (Watts-Strogatz random graph model), and "BA" (Barabási-Albert random graph model).
maxBoot	integer indicating the number of bootstrap resamples (default is 500).
...	Other relevant parameters for <code>graph.param.estimator</code> .

Value

A list containing:

- `parameters`: a vector containing the estimated parameters for each graph.
- `F.value`: the F statistic of the test.
- `p.value`: the p-value of the test.

References

Andre Fujita, Eduardo Silva Lira, Suzana de Siqueira Santos, Silvia Yumi Bando, Gabriela Eleuterio Soares, Daniel Yasumasa Takahashi. A semi-parametric statistical test to compare complex networks, *Journal of Complex Networks*, cnz028, <https://doi.org/10.1093/comnet/cnz028>

Sheather, S. J. and Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *Journal of the Royal Statistical Society series B*, 53, 683-690. <http://www.jstor.org/stable/2345597>.

Examples

```
set.seed(42)
model <- "ER"
G <- list()

# Under H0
G[[1]] <- igraph::sample_gnp(50, 0.5)
G[[2]] <- igraph::sample_gnp(50, 0.5)
G[[3]] <- igraph::sample_gnp(50, 0.5)
result1 <- sp.anogva(G, model, maxBoot = 300)
result1

# Under H1
G[[1]] <- igraph::sample_gnp(50, 0.5)
G[[2]] <- igraph::sample_gnp(50, 0.55)
G[[3]] <- igraph::sample_gnp(50, 0.5)
result2 <- sp.anogva(G, model, maxBoot = 300)
result2
```

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